SWAROOP NATH

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Research Scientist / Engineer — Efficient Inference & Reasoning for Large Language Models

Research Experience

Reflection AI — Member of Technical Staff, Reasoning Post-Training May'26 – Present

- **Drove the development of a synthetic data-generation pipeline** for highly difficult reasoning data, improving the team's math-reasoning benchmark (↑ 12%) to a new internal SOTA.
- Built in-house tools for automated monitoring of reasoning traces in RL training, used by a team of **20+ people**.

Google DeepMind — Researcher Jul'24 – May'26

- **Led the development of an efficient-reasoning algorithm** that reduced verbosity by ∼50% across the Gemini Flash model series while preserving reasoning quality.
  - Catalyzed a **new broader efficient-reasoning workstream** within Gemini's Thinking team.
  - **Established collaboration with Gemini's Thinking team**, helping seed follow-on Thinking research within our team and broadening the impact of the work across Gemini.
- **Led a novel research direction "Interruptible Thinking"** to enable a safe "Answer Now" in Gemini (external coverage); developed a novel decoding-time algorithm that lets a model's reasoning be safely interrupted mid-generation and still return a response grounded in its strongest hypothesis so far.
- **Designed a training-free, clustering-based efficient-attention algorithm** that cut inference latency ∼40% with no loss in generation quality.

LinkedIn — AI/ML Research Intern May'23 – Jul'23

- **Built a context-aware spam-detection model** for LinkedIn comments, incorporating neighboring comment/post context, improving overall pipeline quality ∼20%.

Selected Publications

(* = equal contribution)

- Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities **Technical Report**
  Google DeepMind (large-scale collaboration)
- One Prompt To Rule Them All: LLMs for Opinion Summary Evaluation **ACL'24**
  Swaroop Nath, Tejpalsingh Siledar*, Pushpak Bhattacharyya, et al.*
- Reinforcement Replaces Supervision: Query focused Summarization using Deep Reinforcement Learning **EMNLP'23**
  Swaroop Nath, Pushpak Bhattacharyya, Harshad Khadilkar
- Leveraging Domain Knowledge for Efficient Reward Modelling in RLHF: A Case-Study in E-Commerce Opinion Summarization **arXiv**
  Swaroop Nath, Tejpalsingh Siledar*, Pushpak Bhattacharyya, et al.*
- Transformers are Expressive, But Are They Expressive Enough for Regression? **arXiv**
  Swaroop Nath, Harshad Khadilkar, Pushpak Bhattacharyya

Mentoring

- **Mentored 2 junior master's students** in scaling e-commerce opinion summarization from a few hundred to 1000s of reviews per product, with a linear-cost approach, replacing a quadratic one — arXiv.
- **Mentored students in an academic bootcamp** on various topics in LLMs, facilitating their transition into ML research from other academic backgrounds.

Honors, Recognition & News

- Recipient of **Winifred B. Fernandes Research Excellence Award** for my MS by Research Thesis.
- Recipient of **Microsoft Research India grants** to present my works at ACL and EMNLP.
- **Best Talk Award**, "Efficient Reward Modelling in RLHF," RISC'24, CSE @ IIT Bombay.
- Invited talks/tutorials at **Hyperbots**, **Qualcomm** (RLHF & Alignment), and **ICON'23** (Vision-Language Models).
- Reviewer: ACL, EMNLP, EACL; co-reviewer NeurIPS'24; **Outstanding Reviewer**, ACL'25 (3 of 4 papers).

Education

Indian Institute of Technology, Bombay — MS by Research, Computer Science & Engineering Aug'21 – Jun'24

GPA 9.76/10 (Dept. Rank 2) · Advisors: Prof. Pushpak Bhattacharyya, Prof. Harshad Khadilkar

Thesis: "Task-Aligned Reward Modeling for Reinforcement Learning in Text Summarization"

National Institute of Technology, Allahabad — B.Tech, Mechanical Engineering Jul'15 – May'19